CONSENT FORM

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You are invited to participate in this study aimed at evaluating the usability, accuracy, and effectiveness of a browser extension designed to classify diabetes health-related claims.

Title of Study: Combating Diabetes Misinformation: A Transformer Model Approach

Sponsor of the Study: This study is funded by the **Natural Sciences and Engineering Research Council of Canada (NSERC)**.

Principal Investigator: Ifeoma Adaji, Department of Computer Science, Mathematics, Physics and Statistics, 250-807-8332, ifeoma.adaji@ubc.ca
Co-Investigator: Linda Okpanachi, Department of Computer Science, Mathematics, Physics and Statistics, linda.okpanachi@.ubc.ca

Purpose and Procedure: This study focuses on evaluating a browser extension designed to classify health-related claims, such as those about diabetes, as accurate, exaggerated, misconstrued, or false. The extension aims to assist users in assessing the credibility of health information. By participating in this study, you will help us better understand how users perceive the tool's ease of use, perceived usefulness, intention to use in the future, accuracy and effectiveness, credibility and trust and overall participant satisfaction.

You will be asked to test the browser extension by inputting three to five diabetes-related health claims you may have seen on different social media platforms or may have come across into the provided interface on the extension over a period of one week followed by a questionnaire.

The questionnaire in this study is divided into six sections:

- Section 1: Questions about perceived usefulness.
- Section 2: Questions about ease of use.
- Section 3: Questions about accuracy and effectiveness.
- Section 4: Questions about credibility and trust
- Section 5: Questions about intention to use in the future
- Section 6: Questions about demographics and usage validity of the extension.

The entire process will take approximately 13–**20 minutes** to complete. If you are uncomfortable answering any question, you may close your browser, and the survey will end. While most questions require a response, demographic questions allow you to select "prefer not to answer.

Validation Test: To validate the participation, participants will be required to submit your email address for verification purposes. Your email address will be stored separately from survey responses and will be securely deleted after the follow-up survey is completed**.**

Study Results: Aggregated results from this study will appear in peer-reviewed conferences and scientific journals. The research will contribute to the graduate thesis of the co-investigator, which will be publicly available on **UBC's cIRcle repository**. The data will be retained for five years post-publication on a UBC server and will be deleted afterward.

Potential Risks: We do not anticipate any significant harm from participating in this study. However, we have clearly articulated the following potential risks in the consent form:

- **Privacy Concerns**: Although no personal identifiable information (PII) will be collected through the extension, participants may still be concerned about data privacy. The only data collected will be email addresses for verification and communication.
- **Discomfort with Incorrect Information**: Participants may encounter sensitive or incorrect information while interacting with the prototype. Participants can stop at any time without any

consequences.

- **Frustration with Prototype Bugs**: The prototype is under development, and participants may experience frustration due to incomplete functionality. Pre-testing has been conducted, and a support email will be provided for troubleshooting.
- **Over-reliance on Prototype Predictions**: Participants will be informed that the tool is a **research prototype** and may not always provide accurate results. A disclaimer will be included, encouraging participants to consult credible sources or professionals for accurate health information.

Potential Benefits: Findings from this study will provide insights into the usability, accuracy, and effectiveness of tools for evaluating health information. This research will help guide future development of systems designed to improve access to credible health information.

Data Privacy and Confidentiality: The browser extension will not collect or store any personal data (such as names, browsing history, or interaction data). The extension will also not collect location information or timestamps of use. Your email address (provided for validation purposes) will be stored separately from your survey responses and deleted after the study concludes. The survey is hosted on the UBC-licensed Qualtrics platform, which is FIPPA compliant. All data will be securely stored in Canada and later transferred to the UBC server for long-term storage. Data will be retained for 5 years post-publication and will then be securely deleted. Survey responses will be de-identified for analysis and future research.

Right to Withdraw: Participation in this study is voluntary, and you can decide not to participate at any time by closing your browser. Once submitted, survey responses cannot be removed as they are de-identified and stored separately from email identifiers.

Renumeration: No monetary incentive will be provided for participation in this pilot study. Participation is voluntary, and participants will be contributing to research that supports the development of tools to combat health misinformation.

Contact: Respondents may find out about the results of the study by contacting the researcher.

Complaints: If you have any concerns or complaints about your rights as a research participant and/or your experiences while participating in this study, contact the Research Participant Complaint Line in the UBC Office of Research Ethics toll-free at 1-877-822-8598 or the UBC Okanagan Research Services Office at 250-807-8832. It is also possible to contact the Research Complaint Line by email (RSIL@ors.ubc.ca). **Please reference the study ID H24-03921** to allow staff to better assist you.

Consent to Participate: Participation in this study is voluntary, and you can decide not to participate at any time by closing your browser. By completing and submitting this questionnaire, your free and informed consent is implied and indicates that you understand the above conditions of participation in this study. Once submitted, your survey responses cannot be removed as they are de-identified.

Do you consent to participate in this study?"

- ○ No, I do not agree to participate
- ○ Yes, I agree to participate

Access Instructions

WELCOME TO THE STUDY

Thank you for agreeing to participate in this study!!! To begin, you will need to add the chrome extension to your browser that you will be testing as part of this study.

Steps to Use the DiaBERT Extension:

Click on the Link to Add the Extension to your Chrome

Browser:
[DiaBERT Classifier]

Open the Extension: Once added, click the DiaBERT extension icon which will reveal the extension interface. You need to be connected to the internet to use the extension, as it communicates with a remote server

How to Use the Extension:

1. **Input a Claim**: Type or paste a diabetes health-related claim into the input box provided. You can input text from different social media platforms regarding diabetes
    - Example: “Eating too much sugar causes diabetes.”
2. **Analyze the Claim**: Click the "Classify" button to see the results.

- The extension will classify the claim and provides a confidence score.
- The extension also has an explainability feature that provides explanations as to how the model arrived at its decision.
- You can click the clear button if you want to add another claim
- You can hover the mouse on the label's bar chart to see the confidence score of the model regarding its prediction

3. **Record Your Observations**:

- You will need to record the claims and the results in the survey form provided.

4. **After Testing the Extension:** Return to take this survey

If you encounter any issues installing the extension, please contact us at linda.okpanachi@ubc.ca

Section 1: Perceived Usefulness

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
My ability to identify accurate diabetes-related information would be difficult without DiaBERT.	○	○	○	○	○
Using DiaBERT gives me greater control over the accuracy of the diabetes-related information I consume or share.	○	○	○	○	○
Using DiaBERT improves my ability to evaluate diabetes-related information.	○	○	○	○	○
The DiaBERT extension addresses my needs for identifying diabetes-related misinformation.	○	○	○	○	○
Using DiaBERT saves me time when evaluating diabetes-related content.	○	○	○	○	○
DiaBERT enables me to identify misleading diabetes-related information more quickly.	○	○	○	○	○
DiaBERT supports critical aspects of my information evaluation process.	○	○	○	○	○

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
Using DiaBERT allows me to assess more content for accuracy than would otherwise be possible.	○	○	○	○	○
Using DiaBERT reduces the time I spend on evaluating incorrect or irrelevant information.	○	○	○	○	○
Using DiaBERT enhances my effectiveness in identifying diabetes-related misinformation.	○	○	○	○	○
Using DiaBERT improves the quality of the information I rely on for diabetes-related decisions.	○	○	○	○	○
Using DiaBERT increases my productivity when filtering diabetes-related content.	○	○	○	○	○
Using DiaBERT makes it easier to evaluate diabetes-related information.	○	○	○	○	○
Overall, I find the DiaBERT extension useful for identifying diabetes-related misinformation.	○	○	○	○	○

Section 2: Perceived Ease of Use

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I often become confused when I use the DiaBERT browser extension.	○	○	○	○	○
I make errors frequently when using DiaBERT.	○	○	○	○	○
Interacting with the DiaBERT extension is often frustrating.	○	○	○	○	○
I need to consult the user manual or instructions often when using DiaBERT.	○	○	○	○	○
Interacting with DiaBERT requires a lot of my mental effort.	○	○	○	○	○
I find it easy to recover from errors encountered while using DiaBERT.	○	○	○	○	○
The DiaBERT extension is rigid and inflexible to interact with.	○	○	○	○	○

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I find it easy to get DiaBERT to perform the tasks I want it to do.	○	○	○	○	○
The DiaBERT extension often behaves in unexpected ways.	○	○	○	○	○
I find it cumbersome to use the DiaBERT extension.	○	○	○	○	○
My interaction with DiaBERT is easy for me to understand.	○	○	○	○	○
It is easy for me to remember how to perform tasks using DiaBERT.	○	○	○	○	○
The DiaBERT extension provides helpful guidance in performing tasks.	○	○	○	○	○
Overall, I find the DiaBERT browser extension easy to use.	○	○	○	○	○

Did you encounter any technical issues while using the tool?

- ◯ Yes
- ◯ No

Please describe the issue.

Section 3: Accuracy and Effectiveness

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
The tool provided accurate classifications for the claims I tested	◯	◯	◯	◯	◯
The results displayed by the tool were clear and easy to interpret.	◯	◯	◯	◯	◯
I felt successful in accomplishing the tasks I performed with the tool.	◯	◯	◯	◯	◯
The tool improved my ability to assess the accuracy of diabetes-related claims.	◯	◯	◯	◯	◯

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I found the tool helpful for evaluating diabetes health information.	○	○	○	○	○

Can you describe an example where the tool provided an accurate or helpful result?

Were there any cases where you disagreed with the classification results?

- ○ Yes
- ○ No

Please explain

Section 4:Credibility and Trust

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I believe the tool provides reliable information.	○	○	○	○	○
I trust the results provided by the tool.	○	○	○	○	○
The tool increased my confidence in identifying diabetes misinformation.	○	○	○	○	○
I feel comfortable relying on the tool for evaluating diabetes health-related claims.	○	○	○	○	○
The tool seemed objective and unbiased in its classifications.	○	○	○	○	○

What made you trust or distrust the results provided by the tool?

What improvements would increase your trust in the tool?

Section 5: Intention to Use in the Future

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I intend to use the DiaBERT browser extension regularly in the future.	○	○	○	○	○
I plan to continue using DiaBERT to evaluate diabetes-related information.	○	○	○	○	○
I expect to increase my use of DiaBERT over time.	○	○	○	○	○
I would recommend DiaBERT to others who need to assess diabetes-related content.	○	○	○	○	○

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I am likely to rely on DiaBERT for identifying diabetes-related misinformation in the future.	○	○	○	○	○
I believe DiaBERT will be useful in helping me make better decisions related to diabetes information.	○	○	○	○	○
I am committed to using DiaBERT as part of my process for evaluating online health information.	○	○	○	○	○

Section 6: Demographics and Usage Validity

Age Range:

- ○ 18 - 24
- ○ 25 - 34
- ○ 35 - 44
- ○ 45 - 54

- 55+
- Prefer not to answer

Gender:

- Male
- Female
- Non-binary / third gender
- Prefer not to say

Education

- High school graduate
- Diploma
- Bachelor's Degree
- Master's Degree
- Professional degree
- Doctorate

How often do you interact with health-related information online?

- Never
- Sometimes

- ○ About half the time
- ○ Most of the time
- ○ Always

Please provide examples of three claims you tested in the tool, their classification label and confidence scores.
Example:
Claim: Diabetes can be cured by eating sour grapes
Classification: False
Confidence Scores: 90%

Enter your first claim and the results provided

Claim 1

Classification 1

Confidence Score 1

Enter your second claim and the results provided

Claim 2

Classification 2

Confidence Score 2

Enter your third claim and the results provided

Claim 3

Classification 3

Confidence Score 3

Do you have any additional comments or suggestions about your experience with the tool?

In your own words, how helpful did you find the explanations generated by the system?

Do you think the explanation feature influenced your trust in the system?

- Yes
- No

Why?